02 Mar'26:

→ Checked AI word problems HW

→ Prime factorization to find no. of common factors — see AI notes

HW: Complete AI exercises

→ Percent = / 100

per = divided by

cent = 100

Eg: John has 20 \$. He used 73% of his money. How much is left?

Total money John has $T = 20$ \$

Amount he used $A = 73\% \text{ of } T = \frac{73}{100} \times 20 = 14.6$ \$

Amount remaining $= 20 - 14.6 = 5.4$ \$

→ Ratio / Proportion

A has 23 pizzas. B has 42 pizzas.

Ratio of pizzas of A & B = $A : B = \frac{A}{B} = 23 : 42$

$= \frac{23}{42}$

→ How <u>much</u> percentage did you score if you got 65 out of 70 points?

$x\% = \frac{65}{70}$

$\Rightarrow \frac{x}{100} = \frac{65}{70}$

$\Rightarrow \frac{x}{100} \times 100 = \frac{65}{70} \times 100$

$\Rightarrow x \cong 92.86$

So you scored $x\% = 92.86\%$

→ Ratio of points you scored to the the total points = 65 : 70

= 92.86 / 100

= 0.9286

→ Direct proportion :

If each candy costs 2$,

→ no. of candies $n = 8$; c = cost of candies = $2 \times 8 = 16$$

→ $n = 10$; $c = 20$; $\frac{c}{n} = 2$

→ $n = 33$; $c = 66$; $\frac{c}{n} = 2$

→ $n = 44$; $c = 88$; $\frac{c}{n} = 2$

$c \propto n$ ('c' is directly proportional to 'n')

⇒ $c = kn$ i.e., 'c' increases if 'n' increases

(or) $\frac{c}{n} = k$ (constant) c decreases if n decreases

→ **Indirect proportion :**

If we have to travel a distance 'D',
if our speed increases, time decreases.
or vice versa.

speed $s = 10 \frac{m}{s}$; time = 20s ;

$s = 20 \frac{m}{s}$; time = 10s ;

$s = 40 \frac{m}{s}$; time = 5s .

$s \times t = 200\ m$ is constant

$s \propto \frac{1}{t}$ ('s' is indirectly proportional or inversely proportional to 't')

⇒ $s \times t = k_2$ (constant)

→ See AI notes ; HW: Complete exercises.